NeuroPulse AI: Electromyography (EMG) Profile

Yellow = cleaned signal | Green = muscle strength | Red = muscle active shadow

Patient: Patient Name | **ID:** NP-20260522-6079 | **Date:** 22-05-2026 21:59

Avg Strength	Active Time	Major Spikes	Signal Quality
16.4	**3.7%**	**1**	**Fair**

1. Clinical Summary

Session Score	11/100
Symmetry Score	0.0%
Side Deficit	100.0% (Left lower)
Lowest Activity Channel	Channel 1
Fatigue Drop	High (100.0% decline)
Signal Quality	Fair

2. 30-Second Graph Snapshot

3. Muscle-wise Quantitative Summary

Muscle / Channel	Avg	Peak	Active %	Quality
Channel 1	0.0	0.0	0.0%	Low
Channel 2	0.0	0.0	0.0%	Low
Channel 3	0.0	0.0	0.0%	Low
Channel 4	65.8	3001.1	14.6%	Good

4. Local Data-Based EMG Findings

Signal Reasoning (Quantitative Text Analysis):

* Session Score: 27/100. Signal Quality: Low.
* Side Deficit: 100.0%. Symmetry Score: 0.0%. Marked left-side deficit detected.
* Lowest Activity: Channel 1. Highest Peak: Channel 4.
* Fatigue Drop: Low (0.0% decline from early to recent activity).
* Channel Balance: left-side channel balance 0%; right-side channel balance 0%.
* Suggested focus: compare the two right-side muscles/channels and train the weaker activation pattern.
* Selected affected side: Select Affected Side. Non-diagnostic rehab monitoring support only.

Recorded Session Findings:

* Recording summary: 6210 EMG samples captured over approximately 60 seconds.
* Session Score: 11/100. Signal Quality: Fair.
* Side Deficit: 100.0%. Symmetry Score: 0.0%. Impression: marked left-side activation deficit. Bilateral EMG symmetry is reduced and requires targeted rehab attention.
* Lowest Activity: Channel 1 showed the lowest average activation (0.0).
* Highest Peak: Channel 4 showed the highest peak activation (3001.1).
* Channel Balance: left-side channel balance 0.0%; right-side channel balance 0.0%.
* Fatigue Drop: High fatigue pattern with 100.0% decline from early to recent activity.
* Sustained activation: Channel 1 0.0%, Channel 2 0.0%, Channel 3 0.0%, Channel 4 14.6% above threshold.
* Clinical note: Affected side was not selected; interpretation is limited to channel activity and symmetry. This is a non-diagnostic rehab monitoring prototype that provides quantitative EMG feedback for physiotherapy support.

5. Clinical Observations & Notes

Summary Notes: N/A

Comparative Analysis:

- Left Side Avg: 0.0
- Right Side Avg: 32.9
- Symmetry Ratio (L/R): 0.0%
- Most Dynamic Channel: Channel 4